

Ex. No. 2	ADAPTIVE AND RESPONSIVE UI DESIGN
Date of Exercise	
Git Hub Host Link	

Aim

To create a Flutter application that automatically adapts its layout according to the available screen size.

Description

This experiment demonstrates adaptive and responsive UI design in Flutter using `Layout Builder`. The application displays a `Column` layout for smaller screens such as mobile devices and a `Row` layout for larger screens such as tablets and desktops.

Procedure:

1. Create a new Flutter project using the following commands:

```
flutter create responsive_ui  
cd responsive_ui  
code .
```

2. Run the Flutter application using:

```
flutter run
```

3. Open the `lib/main.dart` file.
4. Use `LayoutBuilder` to check the available screen width.
5. If the screen width is less than 600 pixels, display the mobile layout using `Column`.
6. If the screen width is 600 pixels or greater, display the desktop/tablet layout using `Row`.
7. Add `Card` and `ListTile` widgets to display `Students` and `Courses`.
8. Run the application and resize the browser window or use different emulator sizes.
9. Verify that the layout automatically changes between `Column` and `Row` according to the screen size.

Program

```
import 'package:flutter/material.dart';
```

```
void main() {  
  runApp(const MyApp());  
}
```

```
class MyApp extends StatelessWidget {  
  const MyApp({super.key});
```

```
@override
Widget build(BuildContext context) {
  return MaterialApp(
    title: 'Flutter Demo',
    theme: ThemeData(

      colorScheme: .fromSeed(seedColor: Colors.deepPurple),
    ),
    home: const MyHomePage(title: 'Flutter Demo Home Page'),
  );
}

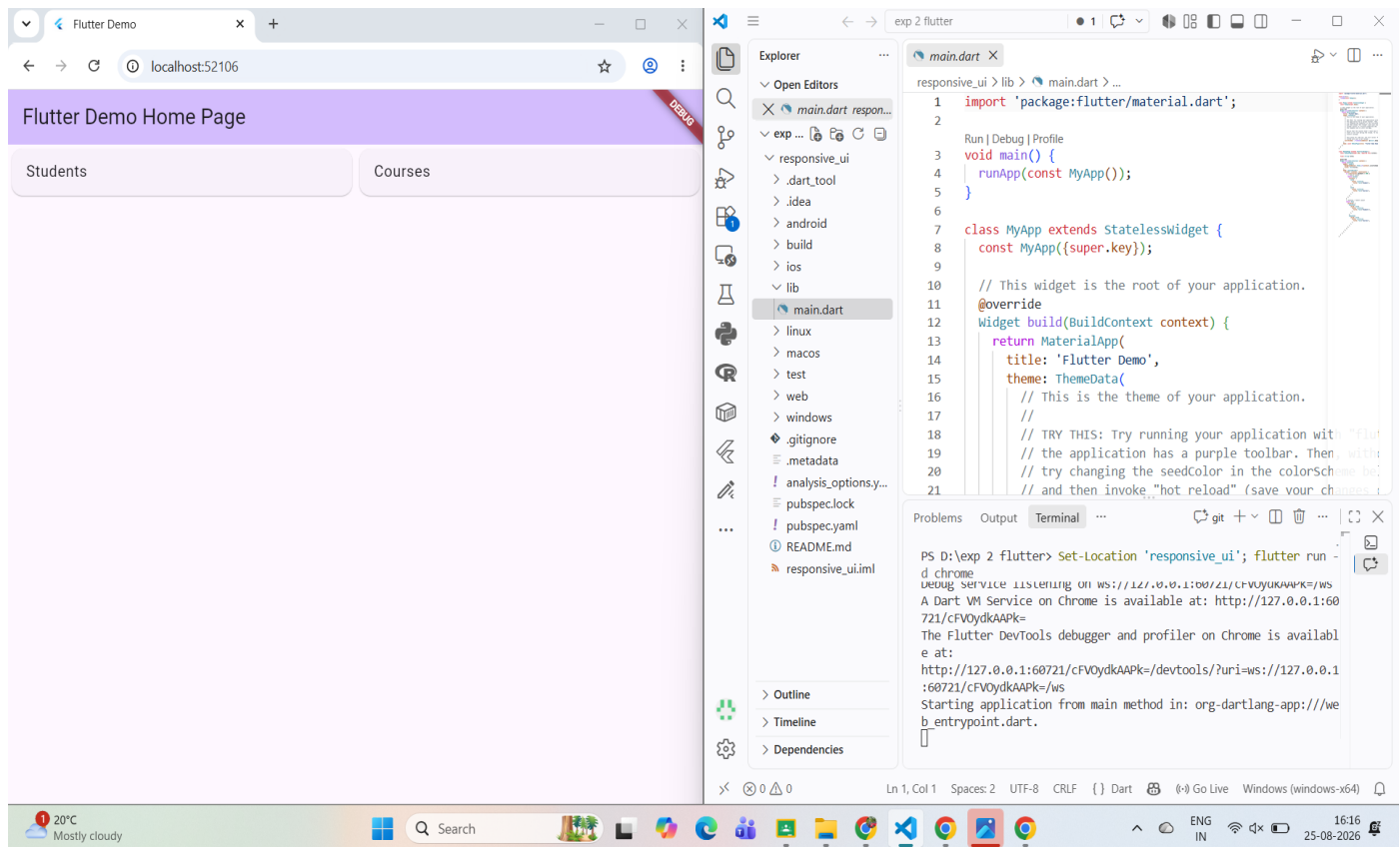
class MyHomePage extends StatelessWidget {
  const MyHomePage({super.key, required this.title});

  final String title;

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        backgroundColor: Theme.of(context).colorScheme.inversePrimary,
        title: Text(title),
      ),
      body: LayoutBuilder(
        builder: (context, constraints) {
          if (constraints.maxWidth < 600) {
            // Mobile Layout
            return Column(
              children: [
                Card(
                  child: ListTile(
                    title: Text('Student'),
                  ),
                ),
                Card(
                  child: ListTile(
                    title: Text('Courses'),
                  ),
                ),
              ],
            );
          }
        }
      );
    }
  }
}
```

```
// Desktop / Tablet Layout
return Row(
  children: [
    Expanded(
      child: Card(
        child: ListTile(
          title: Text('Students'),
        ),
      ),
    Expanded(
      child: Card(
        child: ListTile(
          title: Text('Courses'),
        ),
      ),
    ],
  );
},
);
}
```

Output



Result

Thus, the Flutter application was successfully developed with an adaptive and responsive user interface. The layout automatically changes from Column on smaller screens to Row on larger screens according to the available screen size.